EXHIBIT 32

LUXSHARE TECH

NEXT-GEN POWER MODULES

4:1 1500W IBC

- Delivers 98.5% peak efficiency for energy-optimized power conversion.
- Compact 23 x 17 x 7.7mm design maximizes space efficiency.
- Bottom-side reflow compatibility.

2-Phase Power Block with TLVR

- Achieves 90% peak efficiency for optimized power delivery.
- Ultra-compact 10 x 9 x 4.7mm design enhances density.
- Bottom-side reflow compati
- Ultrafa 300kHz bandwidth
tra t response for dynami

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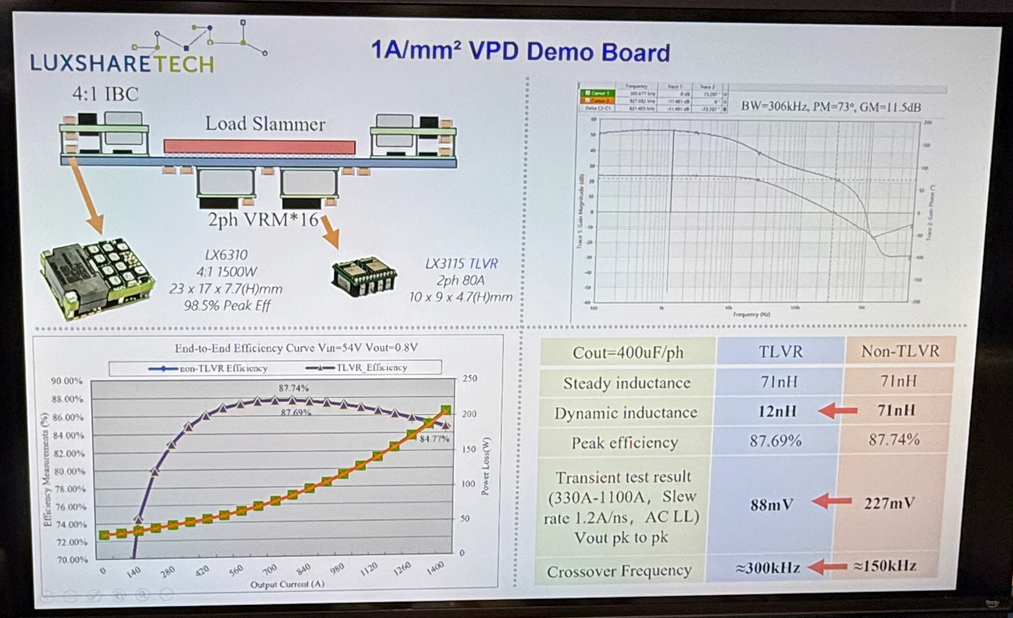
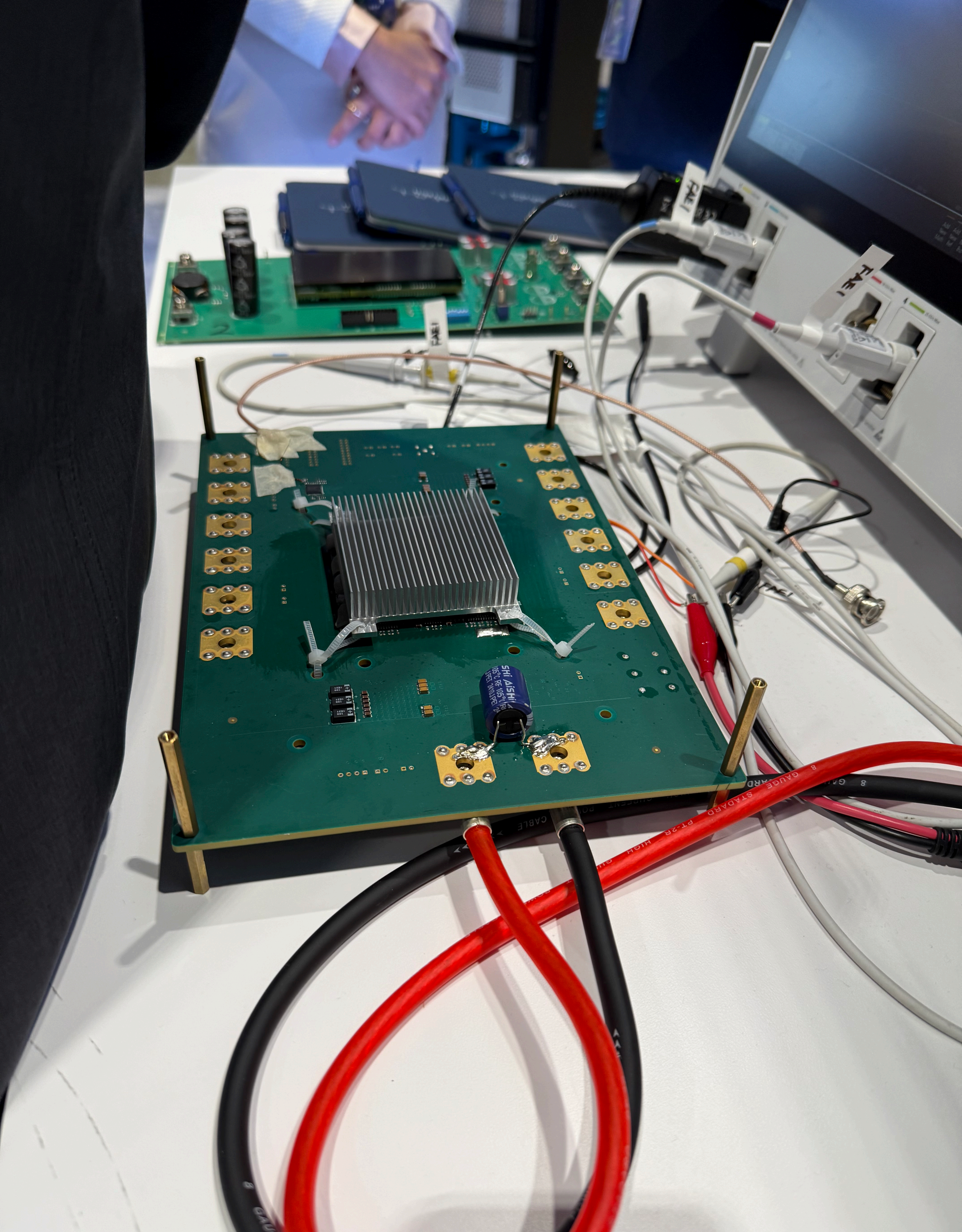


Cout=400uF/ph	TLVR	Non-TLVR
Steady inductance	71nH	71nH
Dynamic inductance	12nH ←	71nH
Peak efficiency	87.69%	87.74%
Transient test result (330A-1100A, Slew rate 1.2A/ns, AC LL) Vout pk to pk	88mV ←	227mV
Crossover Frequency	≈300kHz ←	≈150kHz



CABLE POWER CURRENT HIGH STADARD PT-2R
HIGH CURRENT POWER CABLE
48VIN
GND
GND
12VIN
CLR
SCL
SDA
GND
P2
C975
C973
C972
411319P1PCBV1.1

Vertical Power Delivery for 1000 Amps Machine Learning ASICs

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***Abstract*—The recent explosive growth of machine learning (ML) applications has resulted in a surge of demand for compute performance from custom ASICs, and with the ending of Dennard scaling, power demands of such ASICs have also reached unprecedented levels. In this paper we describe a 48V to point-of-load power solution capable of delivering more than 1000 Amps of current at 0.8V for the latest ML ASICs. The solution follows a two-stage approach, with a fixed-ratio high-density high-efficiency first stage and a second stage module with very high density (1A/mm², 1300W/inch³). The second stage module is integrated into the system board by surface mounting directly underneath the ASIC. This Vertical Power (VPWR) solution removes the "last mile" power loss found in the conventional lateral design, lowers I^2R loss by 70% for 1000 Amps load current, and enables many more other benefits for the overall system such as better signal integrity and thermal performance.**

Keywords—vertical power delivery, high density, voltage regulator modules (VRM), machine learning ASICs, 1000 Amps

I. Introduction

With years of development, the facility level and rack level power solutions of modern data centers are highly efficient, with PUEs as low as 1.1 [1]. However on the tray level PCB board, from the point-of-load (PoL) voltage regulator (VR) to the ASIC load, power is converted to low voltages with very high current and travels for a final lateral distance of a few tens of millimeters with heavily penetrated metal planes, and that's where the majority of power loss happens, as shown in Fig. 1. This IR loss is proportional to the square of current, hence grows much faster than ASIC's own power demand which is only linearly proportional to current. For the latest ML ASICs [2, 3] with lateral power solutions, the distribution power loss alone is estimated to be 5-20% of the ASIC's thermal design power (TDP). This percentage is expected to further increase as ASIC's power scales up. Vertical Power Delivery (VPWR) has been an emerging solution for this lossy "last mile" of lateral distribution [4], as shown in Fig. 2. In the following sections, we present a VPWR solution to convert and deliver more than 1000 Amps of current from directly underneath the ASIC, with 1A/mm² output current density for the vertical module and 70% power loss reduction. With the introduction of the solution, the distribution power loss was reduced from more than 20% of ASIC TDP to below 7%, as shown in Fig. 3.

To authors' best knowledge, this is the first report of a mass-produced, high-density, high-efficiency Vertical Power solution with 1000 Amps output current for ASICs that are not wafer scale packaged.

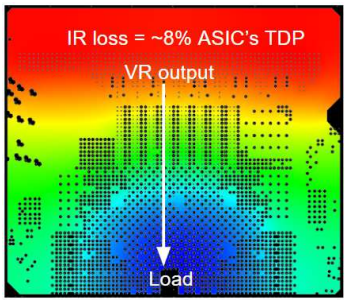

Fig. 1. Distribution loss in lateral power solutions.

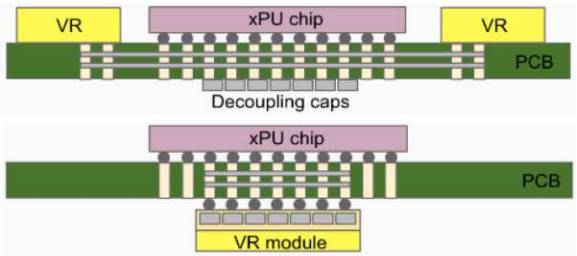

Fig. 2. Lateral power vs. VPWR solutions.

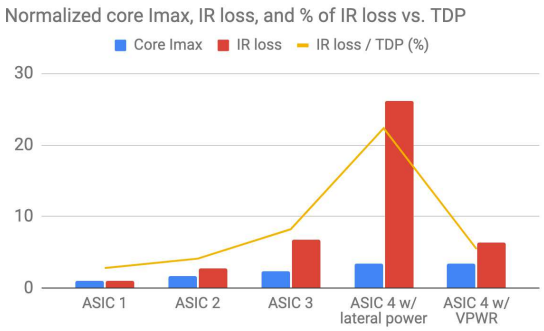

Fig. 3. Distribution power loss vs. chip TDP.

II. Overall Electrical Design

The overall power design architecture follows a two-stage approach [5], with high-efficiency and high-density fixed-ratio converters as the first stage converting 48V tray input power to an intermediate bus voltage, and followed by a second multi-phase Buck stage that converts further down to load voltage range of 0.6-1V, as shown in Fig. 4.

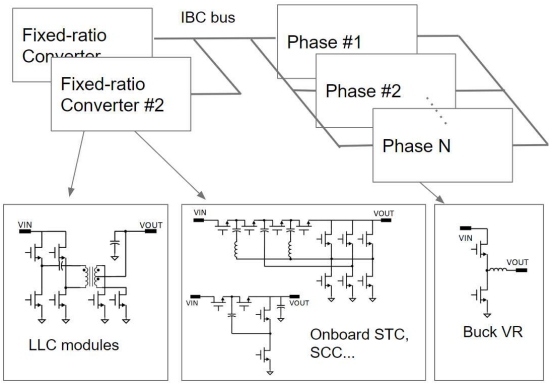

Fig. 4. Two-stage 48V-PoL VPWR architecture.

The intermediate bus voltage level was carefully chosen for the optimal efficiency for the end-to-end system. Key considerations included first-stage efficiency, second-stage power stage efficiency at different input voltages, and the IR loss on the bus plane. Fixed ratios including 10:1 and 8:1 designs were evaluated, and the 8:1 IBC with a 6.75V nominal intermediate voltage was chosen for the final design. The switching frequency of the second stage Buck converter was also carefully studied, considering transient performance requirements, inductor / output caps sizing, and efficiency implications. A switching frequency of 1.2MHz was chosen for the final design. Other techniques such as coupled inductors and trans-inductor VR (TLVR) [6] could be utilized as well if better transient performance is required.

III. Power Module Construction

The first stage IBC converters were placed laterally to the load ASIC, while the second stage VRM was mounted to the bottom of the PCB directly underneath the ML ASIC. A total of 24 phases of power stages were used in the final design, to support 1000 Amp of TDP and 1300 Amp of peak power. Inductors were embedded into the thick PCB underneath the phases. The PCB carrying all the power stages and the inductors was named “power board”. On the other hand, as indicated in Fig. 2, decoupling caps were originally distributed on the bottom side of the main board under the ASIC, to reduce the mid-frequency range power delivery network (PDN) impedance or to mitigate the so-called second droop. In the proposed design, these decoupling caps were absorbed into the second stage VPWR module as well, onto an additional PCB to the power board. This additional “cap board” was then bonded with the power board. The bottom side of the cap board was planted with solder ball arrays that match the ASIC package ball pattern, and the module was then ready to be surface mounted to the main board.

The final second-stage VRM module which is placed vertically, is shown in Fig. 5. The module size is ~33x38x7.5mm, with an output current density of 1A/mm^2 and power density of 1300W/inch^3. Before being placed onto the production board, a test system has been built with a 1000W water-cooled custom load board, as shown in Fig.6. Extensive study has been conducted for not only electrical functionality but also mechanical and thermal properties of the module with custom test vehicles. Weight per ball, BGA ball material, compressive loading capability, warpage, thermal expansion coefficients are among the items that were examined in detail. Reliability and environmental quality testing has also been conducted as well to demonstrate mass production readiness.

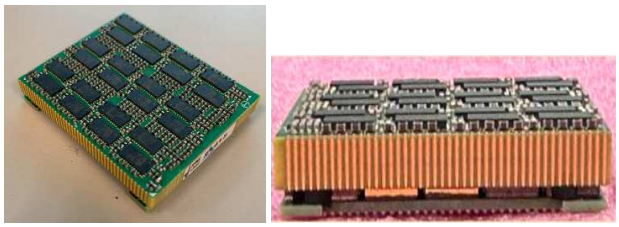

Fig. 5. VPWR module and test board. (a) Top view. (b) Side view.

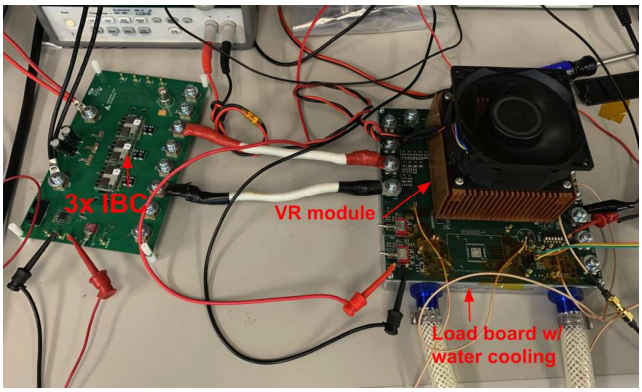

Fig. 6. VPWR test system with custom load board and water cooling.

IV. System Integration

An illustration of the VPWR module’s integration with the ASIC and system board is depicted in Fig. 7, and a photo of assembled VPWR modules with the bottom side cooling assembly is shown in Fig. 8. The VPWR module interacts with the rest of the system in very different ways from the conventional lateral power design. Mechanically the VPWR module impacts the ASIC package and PCB warpage during and after reflow, changing the surface flatness contours of both top and bottom sides. This effect is examined through numerical simulation and correlated with Shadow Moire measurements, as shown in Fig. 9. Thermally, the VPWR module adds one substantial heat source to the vertical stack, but the IR loss underneath the ASIC area is also much lower than lateral power design. With the addition of a bottom side heatsink, it is shown that the ASIC BGA temperature is actually lower than the lateral design. Lower BGA temperature greatly improves the solder ball’s electro-migration capacity,

as simulated in Fig. 10, which is a critical limitation in ultra-high power designs.

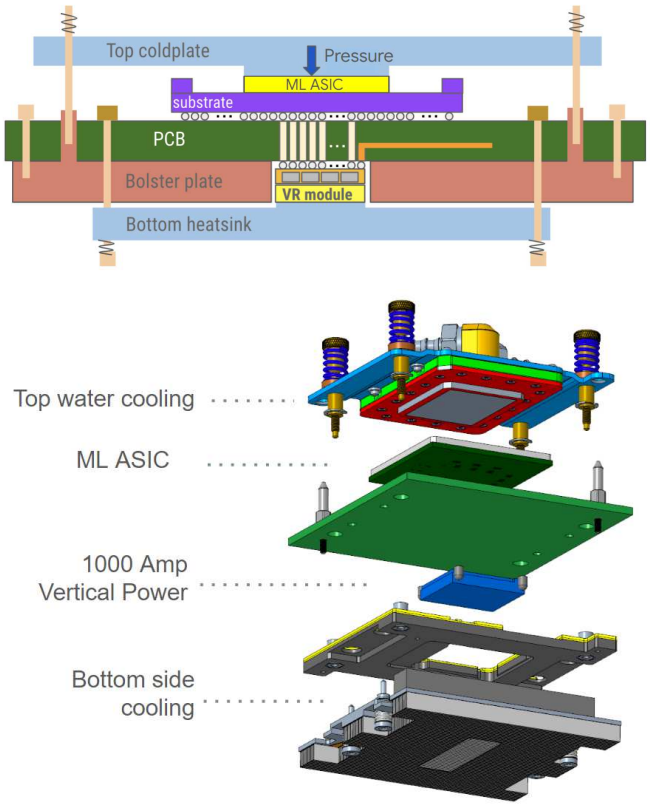

Fig. 7. System integration of VPWR module with ASIC and system board. (a) Side view. (b) Exploded view.

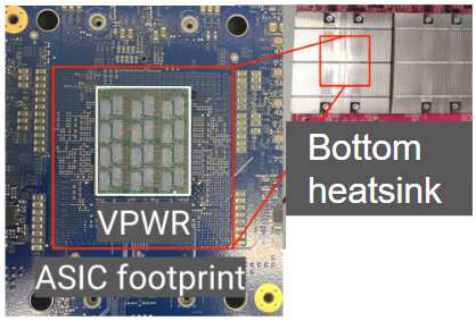

Fig. 8. Assembled VPWR solution with bottom heatsink.

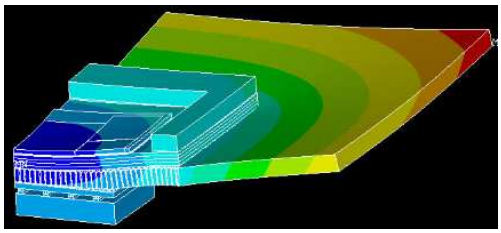

Fig. 9. Structrual analysis of VPWR, PCB and ASIC combined.

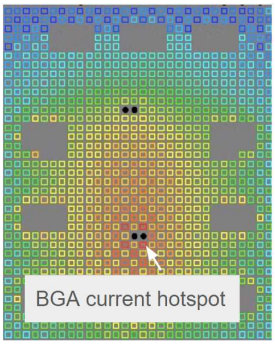

Fig. 10. BGA current heatmap to evaluate solder ball elecro-migration risks.

For manufacturability, the dual side SMT process required careful development. The mixture of via-in-pad (VIPPO) and non-VIPPO vias, combined with the large thermal mass of the VPWR module itself, could easily result in solder hot tear defects on both the ASIC and VPWR module BGA interfaces [7], as shown in Fig. 11. Reworking such modules is challenging, requiring non-standard processes with custom nozzles, compression fixtures, and heat shields, which demonstrated successes in mass production. Finally, the VPWR solution reduced PCB size by roughly 10%, and opened up precious chip peripherals for high-speed SerDes routings. A study has shown that with VPWR solution, the lengths of high-speed PAM4 SerDes channels have been substantially reduced, enabling the use of lower cost fab laminates. The signal routings are also further away from noisy VR output nodes, benefiting high-speed signal integrity in general.

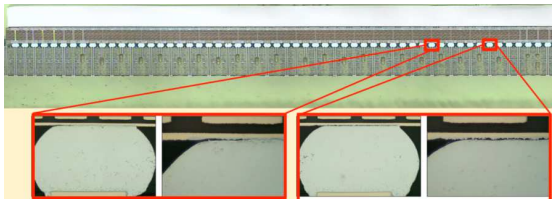

Fig. 11. Solder ball hot tear defects on VPWR and ASIC BGAs.

V. Measurement Results

Power loss, stability, efficiency, and transient performances are simulated and measured on both the test systems and production boards. The load current was generated either by the custom load boards on the test system, or with realistic and synthetic workloads running on the real ML ASIC silicon on the production board. The measurement confirmed IR loss reduction from more than 100W to about 30W for 1000 Amp output current, indicating 70% reduction in IR loss. The 6.75V-to-1V VPWR module efficiency and the overall 48V-to-PoL efficiency curves are plotted in Fig. 12. Note that the latter curve includes all losses on the system board from tray input power to ASIC load, including first and second stages conversion loss, intermediate bus IR loss, output network IR loss, gate driving loss, and everything else. Bode plot measurement was taken with 1000 Amp load current and is shown in Fig. 13, with 68 degree of phase margin and 13.5dB of gain margin. Transient performance was measured with an

800A load step on both test systems with load boards and production boards with real ASICs. Undershoot was 60mV and 68mV respectively, and overshoot was 70mV and 92mV respectively. Note that the load step with the real ML ASIC and workloads is much faster than the bandwidth of the VR, so the measured performance reflects the performance of the full PDN and VR combined, as shown in Fig. 14.

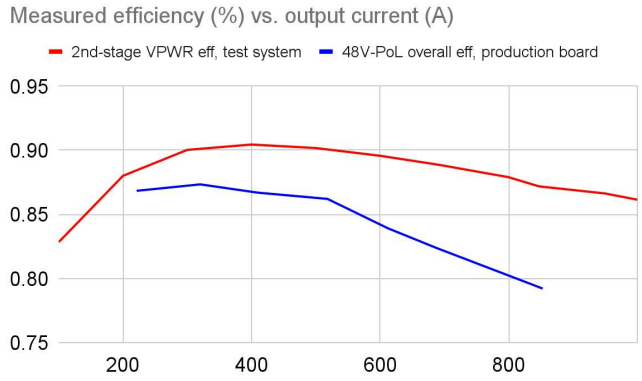

Fig. 12. Measured VPWR efficiency and overall 48V-to-PoL efficiency.

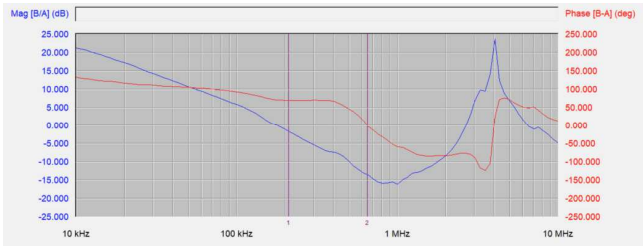

Fig. 13. Measured Bode plot of production boards.

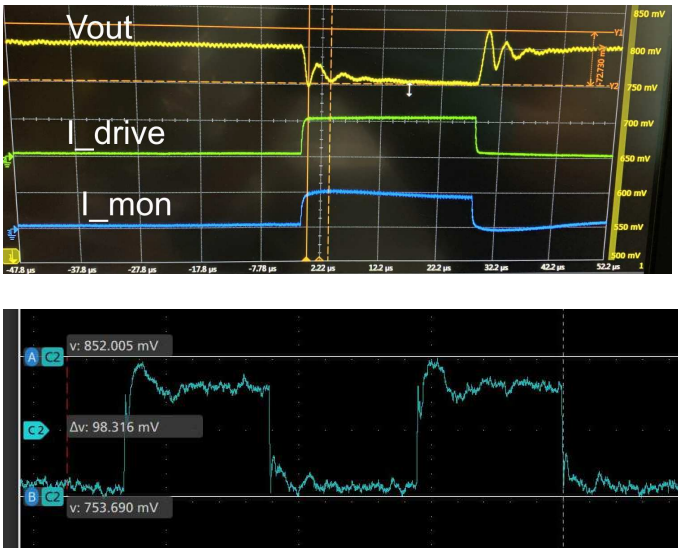

Fig. 14. Measured transient performance of 800A load steps on (a) test boards with custom load board, and (b) production boards with ASICs.

VI. Conclusions and Future Work

In this paper we presented for the first time a mass-produced, high-density, high-efficiency Vertical Power solution with 1000 Amp output current, which combines many of state-of-the-art design, manufacturing and assembly technologies to enable ultra-high power ML ASICs. Our study has shown that ASICs with more than 600A current could reach the inflection point where IR loss in the lateral distribution path exceeds 10% of the chip's TDP. We believe that other than the latest ML ASICs, many chips in other applications are also rapidly approaching this point, including network switches and high-performance CPU/GPU chips. VPWR solution is beneficial for all these cases, not only by substantially reducing the distribution power loss, but also by enabling better system floor plan, improving signal integrity channels, and better managing thermal performance. The challenges are to understand the mechanical interactions and reliability implications with the complex vertical stack, and to ensure sufficiently wide process windows for robust manufacturing and assembly processes.

Acknowledgment

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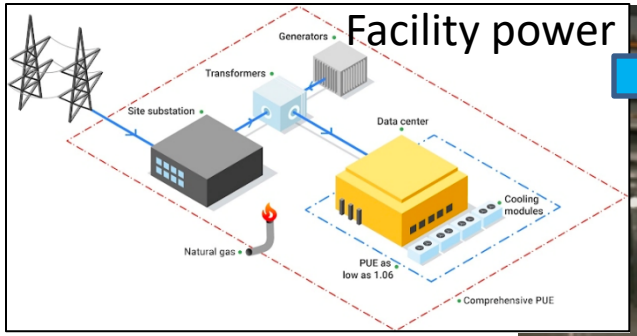
Google, LLC

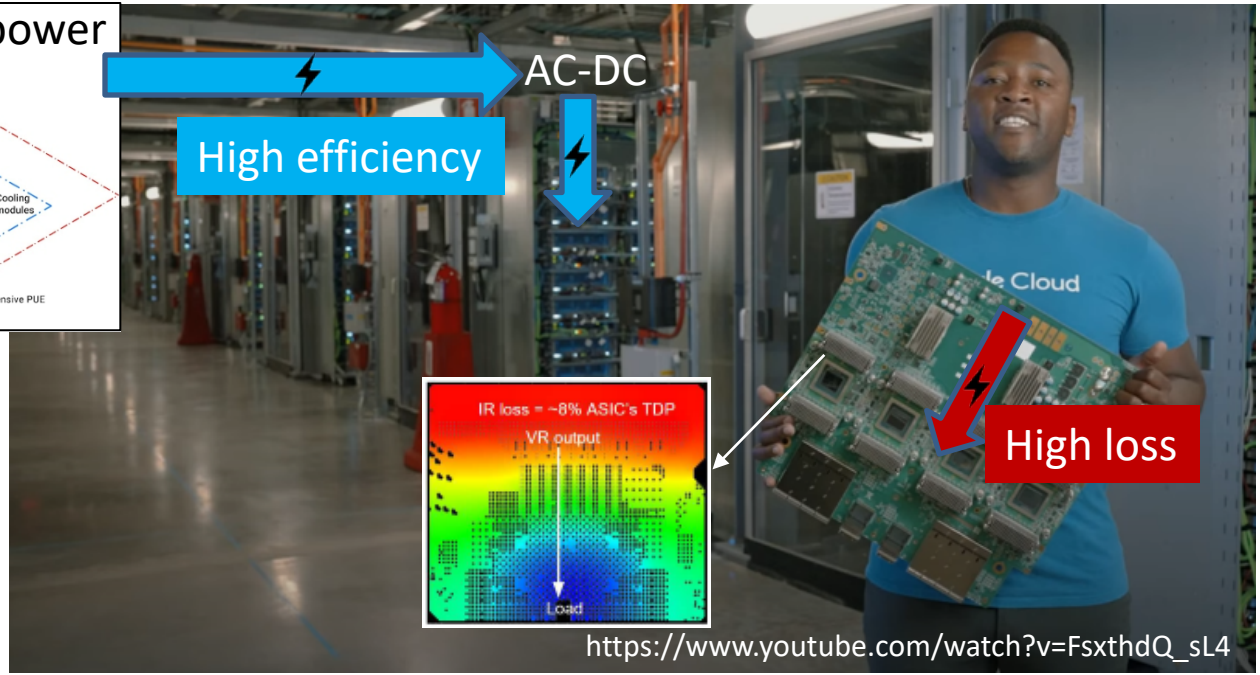
Google

AI Infra Is Expensive

ML fleet TCO breakdown

Lossy “Last Mile” in Power Delivery



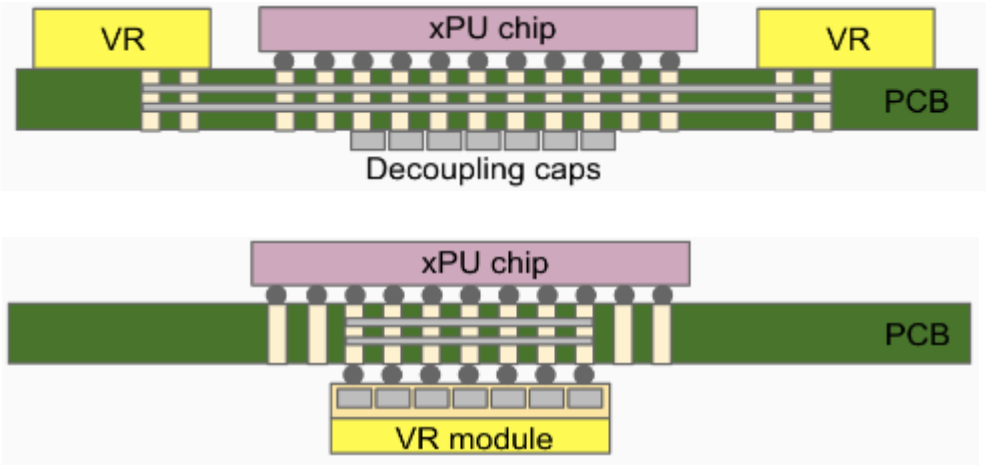


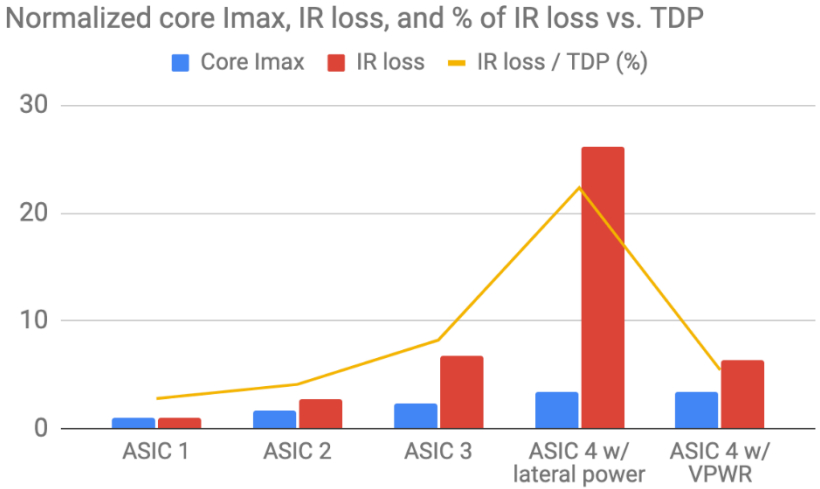
Lossy “last mile” - majority of power loss (sometimes >10% of chip TDP) is from the PCB/package

What is Vertical Power Delivery?

Vertical Power (VPWR):

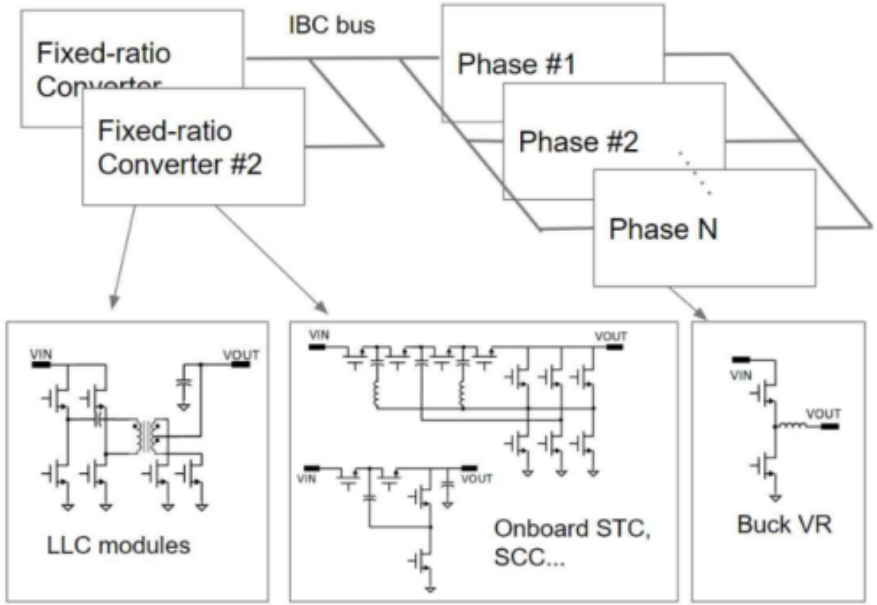
- Delivering 1000 Amp to ASIC in z-dimension
- ~70% reduction in distribution power loss
- Many side benefits (e.g., substantially shorter SerDes channels)





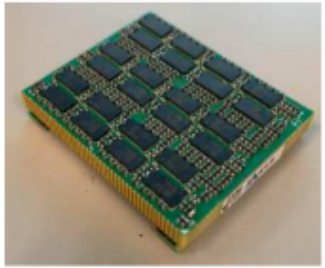
Electrical Design

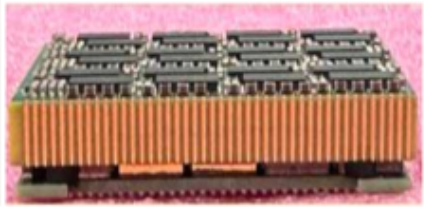
- Two-stage 48V-PoL VPWR architecture
  - 8:1 IBC with 6.75V nominal intermediate voltage as first stage
  - Buck converter at 1.2MHz switching frequency as second stage
  - Coupled inductors and trans-inductor VR topologies compatible

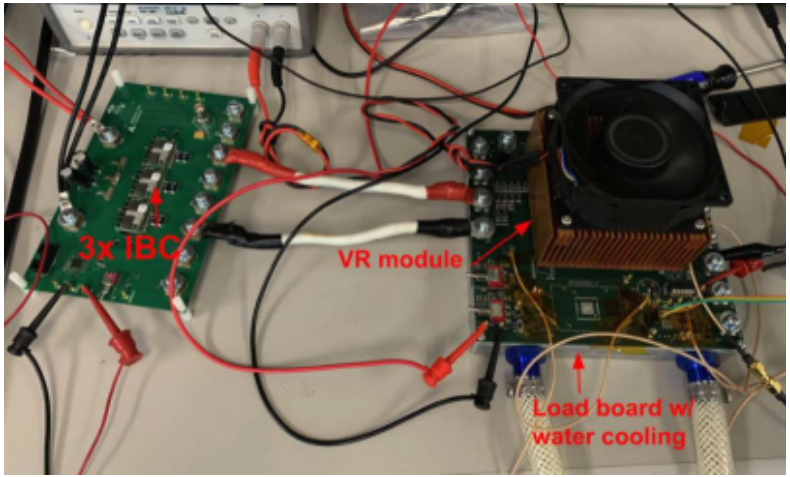


VPWR Module Construction

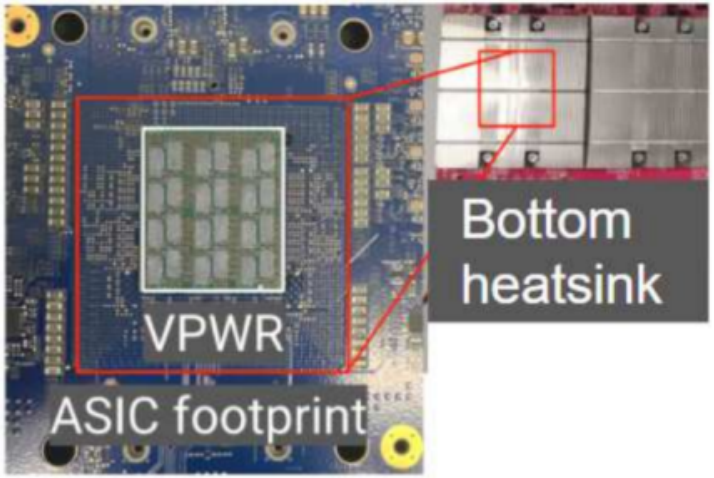
- Module size: ~33x38x7.5mm
- 1000A TDC, 1300A EDC
- High power density: 1A/mm^2, 1300W/inch^3
- Dual-layer PCB structure
  - 24-phase power board
  - High-density cap board
- Test system with 1000W custom water cooled load board



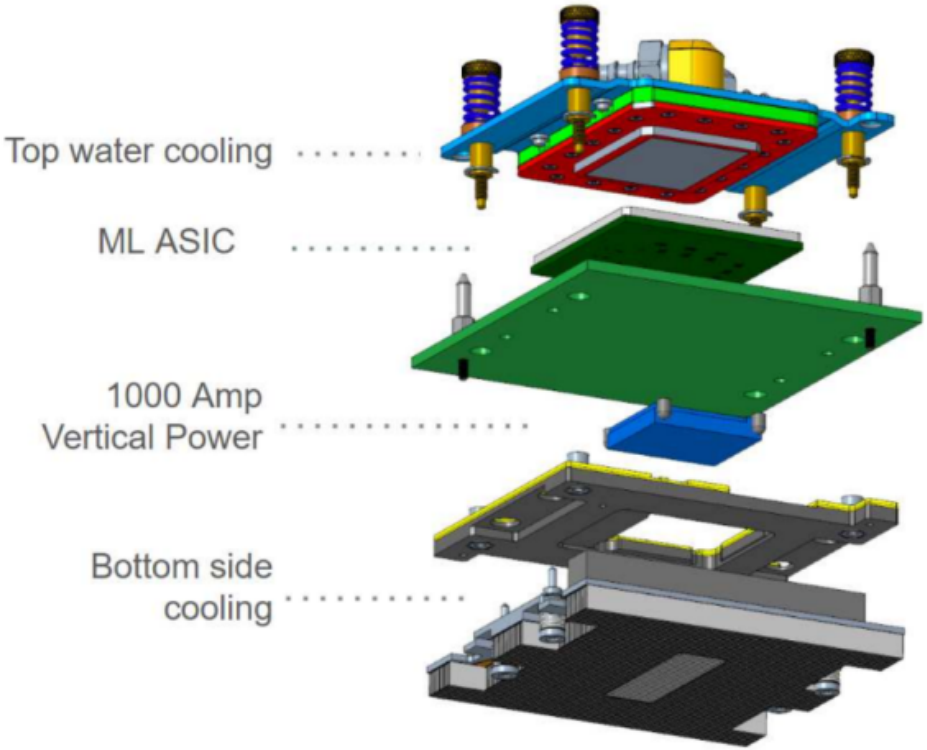




System Integration

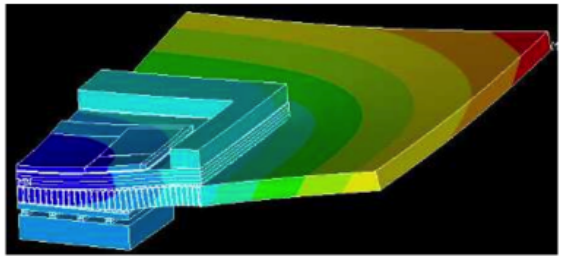


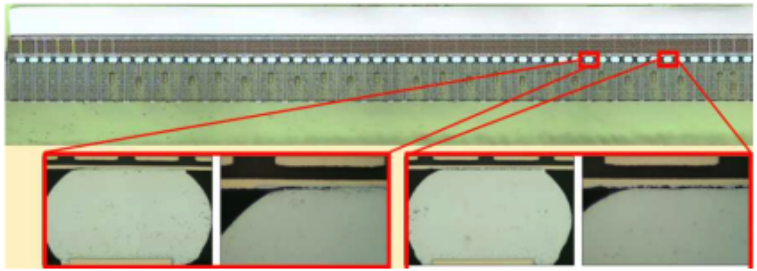
Picture of VPWR on production boards

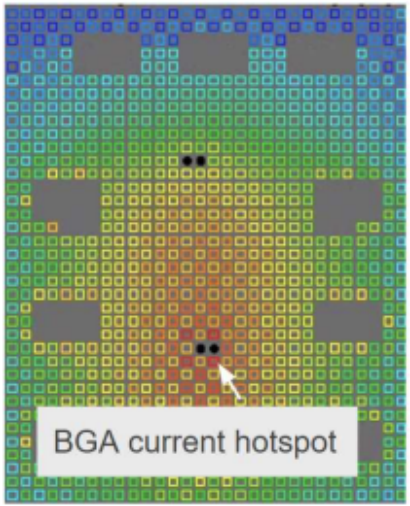


System Integration

- Unique challenges in all aspects of system integration
  - Mechanical design and structural integrity
  - Thermal impact
  - Power integrity, BGA EM
  - Dual side BGA assembly, weight, reworkability

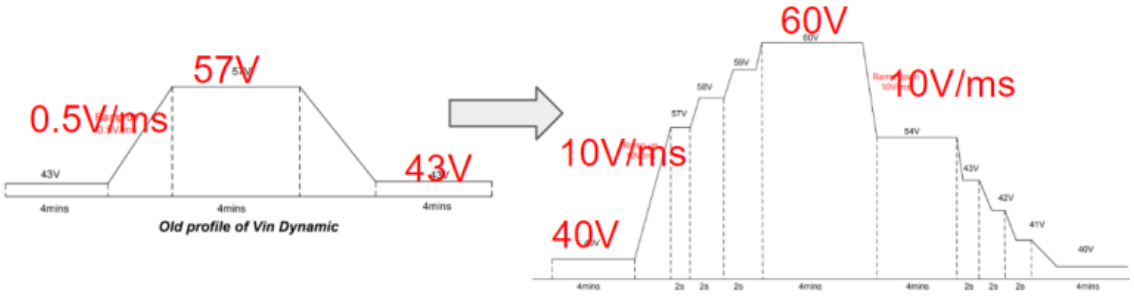






Quality and Reliability

- Mass production ramp-up and readiness
  - Quality assurance on production line
  - Customized burn-in fixtures and systems
    - Much more stressful than typical power module burn-in
    - High slew-rate Vin/load shmoo, TCT, PMBus, etc.
  - Enhanced ORT (on-going reliability test)

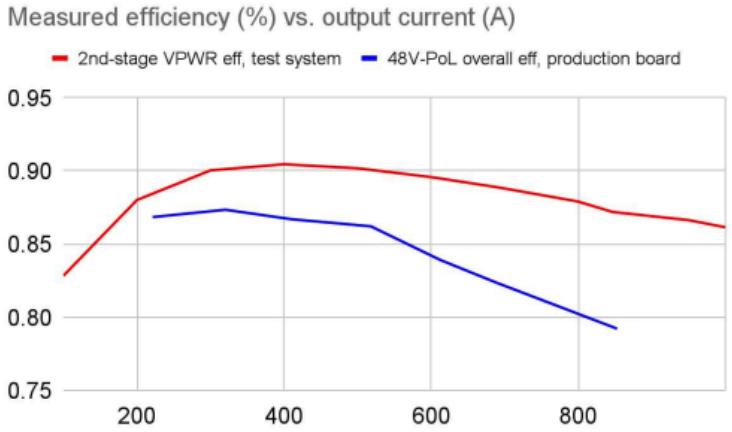


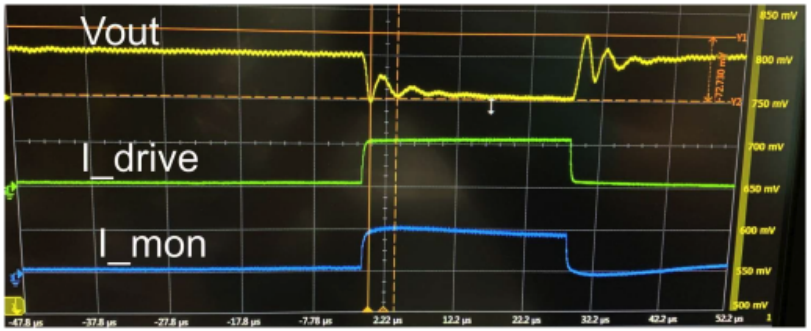
Vin + Load sweep in burn-in chambers

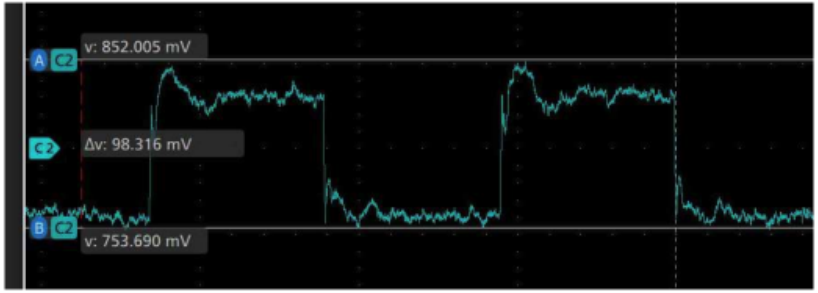
Custom ORT fixture

Measurement Results

- Industry's first 1000 Amps VPWR solution in volume production
  - 87% 48V-to-PoL overall peak efficiency
  - >80% 48V-to-PoL overall efficiency at 800A
  - 68mV undershoot with 800A step







Vertical Power Roadmap

- First report of 1000-Amp VPWR solution in mass production
- Multiple generations of VPWR solutions in production or development
- Advanced module design
  - Multi-rail integration, controller integration
  - Advanced topologies such as TLVR for superior transient performance
  - Modular design and scalability
  - Advanced power management features
- Next-generation solution for ultra-high power ASICs
  - Multi-module design
  - Loss reduction on intermediate bus
- Enhancement in structural integrity, ease for manufacturing, and quality / reliability